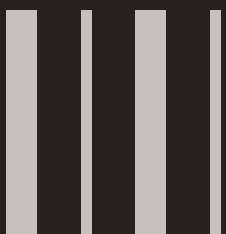


ANDY CANO-AVILA

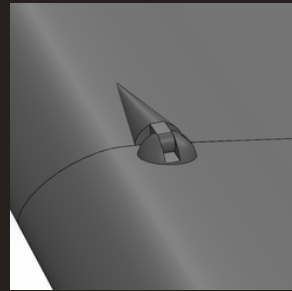
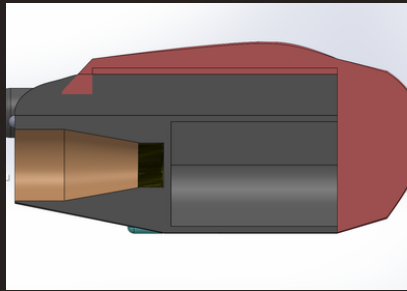
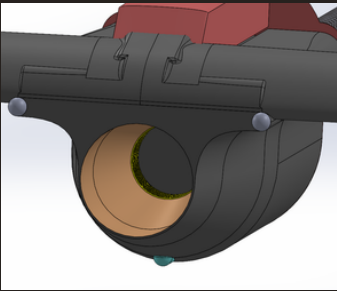
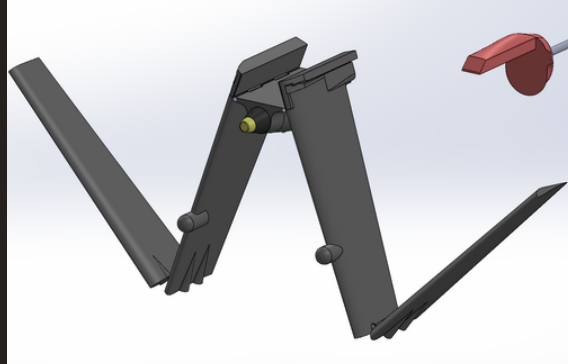
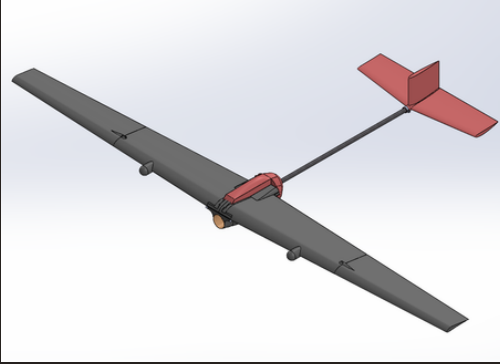
PORTFOLIO

MECHANICAL DESIGN

2023-2026

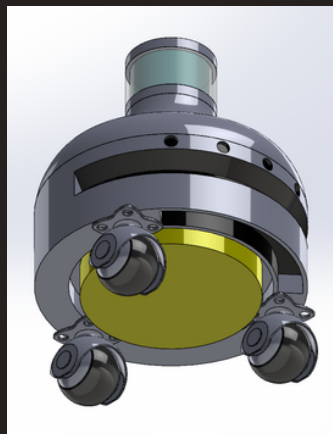
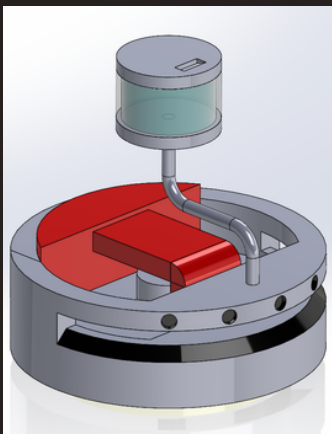
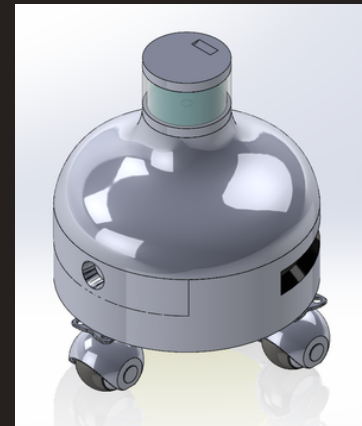
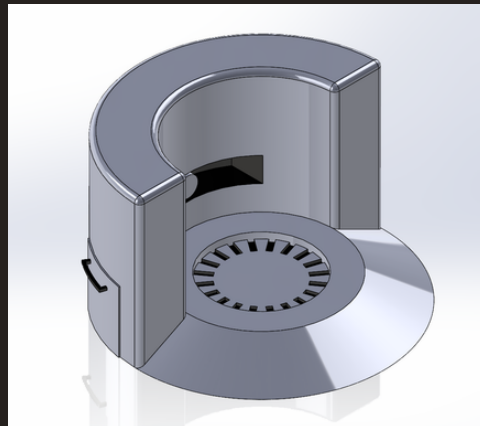
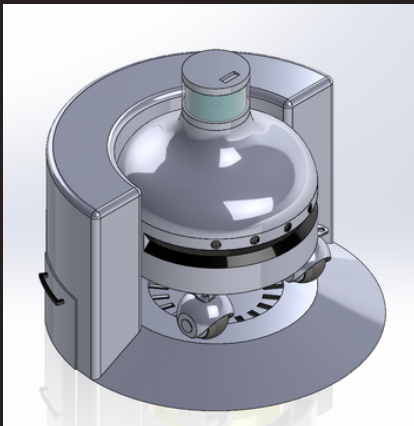


INTERNSHIP WORK



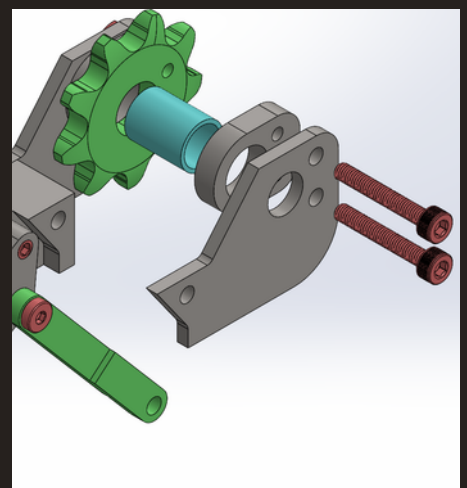
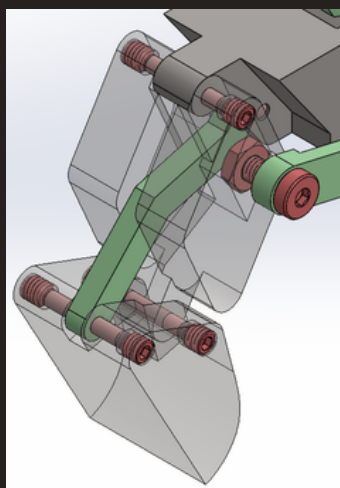
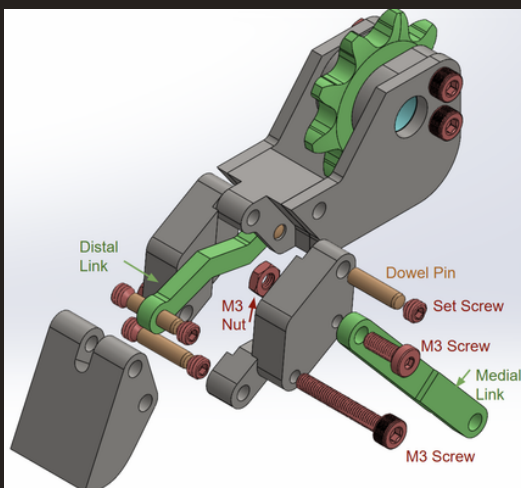
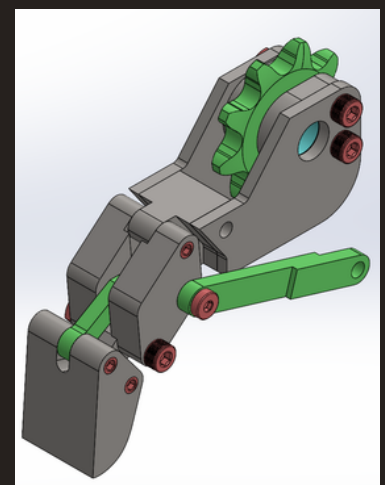
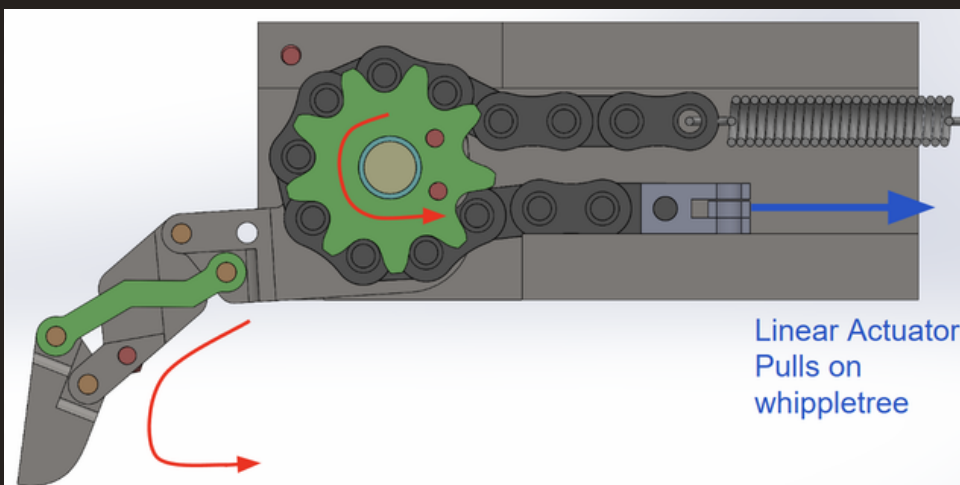
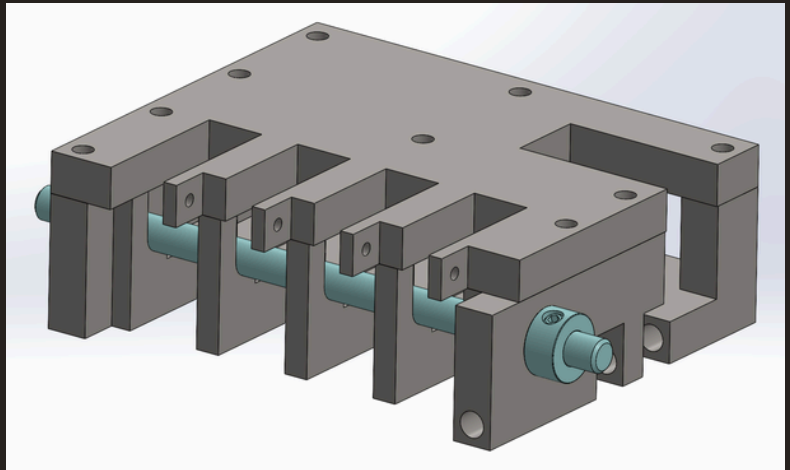
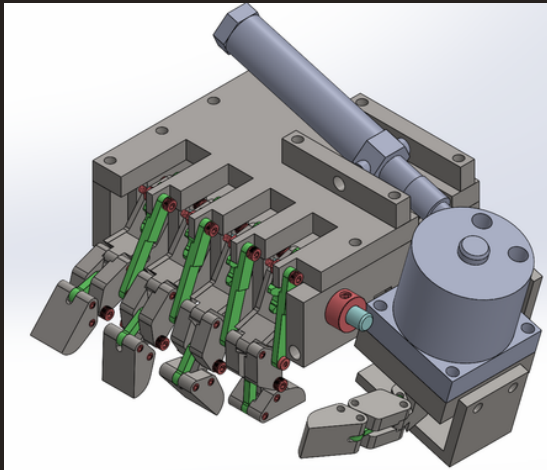
- DRONE DESIGNED WITHIN 36 HRS FROM SCRATCH
- INITIAL CONCEPT DESIGN
- AERODYNAMICS AT THE FOREFRONT
- COMPACTIBILITY
- DEPARTMENT OF DEFENSE STANDARDS
- PROTOTYPED VIA INDUSTRIAL 3D PRINTERS

CLEANING ROBOT



- AUTONOMOUS COUNTERTOP CLEANING ROBOT
- AIDING THE LIVES OF PEOPLE IN A SUSTAINABLE WAY (EASE OF USE)
- CHARGING & UNLOADING STATION
- LIDAR / SUCTION / SPRAY / SPONGE
- 24-HOUR DESIGN WINDOW
- 12X12" FOOTPRINT
- MATERIAL AND ELECTRONICS INTEGRATION
- LIFESPAN AND MARKET SUCCESS

LANL CAPSTONE TEAM



- HYDRAULIC ACTUATED ROBOTIC HAND - 1 DOF FINGER & 2 DOF THUMB
- ROBUST & RUGGED FOR HAZARDOUS WASTE CLEANUP INITIATIVES
- DESIGNED TO UTILIZE POWER/HAND TOOLS
- WHIPPLETREE FORCE DISTRIBUTION
- 4 TOTAL HYDRAULIC ACTUATORS OPERATING AT 400 PSI
- 100+ HOURS DESIGNING FROM SCRATCH
- ISO/ANSI TOLERANCING
- FEA STRESS & DEFLECTION ANALYSIS
- SROCKET-CHAIN & 2-BAR LINKAGE PASSIVE JOINTS FOR POWER TRANSMISSION
- DESIGNED TO BE FABRICATED IN-HOUSE VIA MILLING, CNC, & WATER JET